

GNSS-IR Reflector Height Estimation – Technical Report

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1. Introduction

Global Navigation Satellite Systems (GNSS) signals are not only used for positioning but also for environmental sensing.

When GNSS signals are received, part of the signal arrives directly from the satellite, while another part is reflected from nearby surfaces such as the ground.

This reflected signal causes **multipath interference**, which appears as oscillations in the Signal-to-Noise Ratio (SNR).

GNSS-IR (GNSS Interferometric Reflectometry) exploits these oscillations to estimate properties of the reflecting surface, such as **reflector height**.

2. Objective

The main objective of this work is:

-  To estimate the height of the reflecting surface (ground) using GNSS SNR data.
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3. Data Used

Data Type	Description
RINEX Observation File	.24o (GPS only)
Precise Orbit File	SP3
Satellite System	GPS (G)

◆ 4. Methodology

The complete processing pipeline includes the following steps:

◆ 4.1 Satellite Position & Geometry

- Satellite positions obtained from SP3 files
- Receiver position converted from **ECEF** → **Geodetic**
- Computation of local **ENU coordinates** (East, North, Up)

◆ 4.2 Elevation Angle Calculation

For each satellite and time:

$$\text{elevation} = \arctan(\text{up} / \sqrt{\text{east}^2 + \text{north}^2})$$

◆ 4.3 Satellite Selection

Only satellites passing through:

- $5^\circ < \text{elevation} < 25^\circ$

Additional filtering:

- Number of valid SNR points > 1000

✓ This ensures:

- Strong multipath signal
- Long observation window
- High data quality

◆ 4.4 SNR Processing

- Extract SNR values for each satellite
- Remove invalid (NaN) data
- Apply filtering based on elevation range

◆ 4.5 Detrending

Polynomial fitting (degree = 3) used to remove the slow trend:

$\text{SNR_detrended} = \text{SNR} - \text{trend}$

✓ Leaves only the oscillations caused by multipath (the desired signal)

◆ 4.6 Transformation to $\sin(\text{Elevation})$

$\text{sin_elev} = \sin(\text{elevation})$

✓ Critical step: GNSS-IR signal is periodic in **$\sin(\text{elevation})$** , not elevation itself.

◆ 4.7 Sorting

Data is sorted according to **$\sin(\text{elevation})$** .

✓ Ensures uniform spacing for FFT analysis.

◆ 4.8 FFT Analysis

- Fast Fourier Transform applied to the detrended SNR signal
- Frequency spectrum computed
- Dominant peak identified

◆ 4.9 Height Estimation

Using the formula:

$$h = (\lambda \times f_{\text{peak}}) / 2$$

Where:

- $\lambda = 0.19029367$ m (GPS L1 wavelength)
- f_{peak} = dominant frequency (cycles per $\sin(\text{elevation})$)

◆ 4.10 Multi-Satellite Averaging

- Heights computed per satellite
- Outlier removal applied:

$$|h - \text{mean}| < 2 \times \text{std}$$

◆ 5. Results

Parameter	Value
Average Reflector Height	1.99 m
Standard Deviation	0.16 m

✓ Consistent with the actual antenna height above the reflecting surface (approx. 2 m)

◆ 6. Data Visualization

Histogram

- Shows distribution of estimated heights from all satellites
- Most values concentrated around 2 meters
- No significant outliers observed

Per-Satellite Plot

- Each satellite gives similar height values
 - Minor variations exist due to noise and satellite geometry
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◆ 7. Analysis & Interpretation

Key Observations

- Most satellites estimate height ≈ 2 m
- Small variation between different satellites
- Stable FFT peak extraction
- No abnormal or extreme values

Interpretation

- The antenna is **approximately 2 meters** above the reflecting surface
 - Multipath signal is **strong and consistent**
 - The processing pipeline produces **reliable and repeatable results**
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◆ 8. Conclusion

✅ A **complete GNSS-IR processing pipeline** was successfully implemented.

Key achievements:

- Accurate reflector height estimation
- High stability (low standard deviation: ± 0.16 m)
- Reliable multi-satellite analysis

✅ Final Result:

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Reflector Height ≈ 2.00 m (± 0.16 m)

◆ 9. Future Work

Possible next steps and improvements:

- Use **multi-constellation** (GLONASS, Galileo, BeiDou)
 - Apply **weighted averaging** based on signal quality
 - Extend to real-world applications:
 - **Soil moisture monitoring**
 - **Snow depth estimation**
 - **Water level detection**
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📎 Attachments

1. Histogram of estimated heights (per satellite)
 2. Per-satellite height plot
 3. Python code used for the full pipeline
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